TECHNICAL REPORT · ROMAN URDU · AASAN ZABAAN

Power Monitor Dashboard Mukammal Samjhauti

Bijli ka Monitoring System — AOR / GE(N) Maint Isld

> Is report me poora dashboard aasan alfaz me samjhaya gaya hai. Har tab kya dikhata hai, har number ka kya matlab hai, kaun sa data asli hai aur kaun sa banaya hua — sab kuch is me hai.
>
> Ye report un logon ke liye hai jo pehli baar ye dashboard dekh rahe hain, aur unke liye bhi jinhen kisi doosre ko samjhana hai.

SYSTEM KA NAAM	**Power Monitor — Electricity Management Dashboard**
PROJECT	**Power_Consumption_Dashboard**
DATA KA SOURCE	**AOR - GE (N) Maint Isld.docx**
KITNE TABS	**12 screens · 110 monitored jagahen**
REPORT KI TAREEKH	**4 September 2026**

Fehrist (Contents)

1	Ye Dashboard Kya Hai	2
2	Bijli ke Basic Alfaz	4
3	Har Tab Kya Dikhata Hai	6
4	Alarm System Kaise Kaam Karta Hai	13
5	Data Kahan Se Aata Hai	14
6	Dashboard Kaise Parhein	15

BAB 1

Ye Dashboard Kya Hai

Sabse pehle ye samajhna zaroori hai ke ye cheez hai kya, aur kis masle ka hal hai.

1.1 Aasan lafzon me

Power Monitor aik website jaisa system hai jo batata hai ke **poore complex me bijli kahan aur kitni kharch ho rahi hai**. Ye complex 360 acre ka hai, is me 2,157 buildings hain, 3 bijli ke substation hain aur 109 transformers hain.

Dashboard par **110 jagahen** monitor hoti hain. Har jagah ka apna code hota hai — jaise `TW-01` (Tube Well No.1) ya `MED-HOSP` (PNS Hafeez Hospital). Har jagah ki aik **limit** set hai. Jab koi jagah apni limit se zyada bijli kharch kare, to dashboard turant **alarm** deta hai, aur wo alarm tab tak screen par rehta hai jab tak koi banda usay dekh kar "Ack" (acknowledge) na kar de.

1.2 Ye kis masle ko hal karta hai

Aise bare complex me pehle sirf ye pata hota hai ke *kya kya cheezen maujood hain* — 980 street lights, 84 motor pumps, 14 tube wells. Lekin teen sawalon ka jawab nahi milta:

- **Bijli ja kahan rahi hai?** Kaun se zone aur kaun si building sabse zyada le rahi hai.
- **Abhi koi kharabi to nahi?** Agar koi pump normal se double bijli le raha hai to uska bearing kharab ho raha hai — aur ye kisi ko pata nahi chalta jab tak wo jam na ho jaye.
- **Nayi building lagane ki gunjaish hai?** Kya substation me itni capacity bachi hai.

Dashboard ye teenon sawalon ka jawab har waqt deta rehta hai.

1.3 Sabse zaroori baat — REG aur SIM

> YE PARHE BAGAIR KOI NUMBER ISTEMAL NA KAREIN
>
> Jo document is project ki bunyaad hai (`AOR - GE (N) Maint Isld.docx`), wo aik **asset register** hai — yaani us me sirf ye likha hai ke kya kya maujood hai aur kitni tadaad me. Us me **bijli ki koi reading nahi hai**.
>
> Hum ne document me search kiya: *voltage*, *frequency*, *power factor*, *kWh*, *meter*, *generator* — in me se aik lafz bhi document me nahi hai. Sifar (zero) natija.
>
> Isliye dashboard par jo **live numbers** chal rahe hain wo asli meter se nahi aa rahe — wo **banaye gaye (simulated)** hain. Lekin structure asli hai.

BADGE	MATLAB	IS ME KYA KYA AATA HAI
REG	Register se aaya	2,157 buildings, 980 lights, 84 pumps, 109 transformers (18,530 kVA), 3 substation, 8 breaker, 475 DB, tube wells ke naam aur jagah. **Ye sab asli hai.**
SIM	Banaya hua	Har live reading — kW, voltage, current, frequency, power factor, energy, breaker ki halat. **Ye asli measurement nahi hai.**

Table 1.1 — Dashboard par har number ke saath chhota sa badge lagta hai jo batata hai ke wo kahan se aaya.

> AIK LINE ME KAISE BATAYEIN
>
> "Structure bilkul asli hai aur hamare register se milta hai. Live numbers abhi model (simulation) ke hain, aur wo saaf saaf mark kiye gaye hain. Jab asli meter lag jayenge to sirf data badlega, dashboard wesa ka wesa chalta rahega."

1.4 Kaun log istemal karenge

KAUN	UNHE KYA CHAHIYE
Control room operator	Har waqt dekhta rahe ke koi kharabi to nahi. Overview aur Alarms tab.
Maintenance engineer	Kisi aik jagah ki tafseel — uska trend, limit, purane alarm. Load Monitoring tab.
Electrical engineer	Voltage, phase imbalance, substation ka load. Voltage & Freq, Power Flow tab.
Facilities manager	Kitni bijli kharch hui, kahan sabse zyada, kharcha kam kaise ho. Energy aur Reports tab.
Planner	Nayi building ke liye capacity hai ya nahi. Forecasting tab.

BAB 2

Bijli ke Basic Alfaz

Ye words dashboard par baar baar aate hain. Pehle aasan matlab, phir thoda technical.

2.1 Power aur Energy — sabse zyada confusion yahan hoti hai

> AIK MISAAL SE SAMJHEIN
>
> **Power (kW)** gaari ke *speedometer* jaisa hai — abhi is waqt kitni tez bijli istemal ho rahi hai.
>
> **Energy (kWh)** gaari ke *meter (odometer)* jaisa hai — ab tak kitni bijli total istemal ho chuki hai. **Bill isi ka aata hai.**

```
Energy = Power × Time

Misaal: aik pump 10 kW ka hai aur 5 ghante chala.
        Energy = 10 kW × 5 ghante = 50 kWh
```

2.2 Zaroori units

UNIT	POORA NAAM	AASAN MATLAB
kW	Kilowatt	Abhi kitni tez bijli chal rahi hai. Aik kettle taqreeban 2 kW ka hota hai.
MW	Megawatt	1 MW = 1,000 kW. Poore site ka total isi me likha jata hai (taqreeban 4.5 MW).
kWh	Kilowatt-hour	Kitni bijli kharch hui. **Bill isi par aata hai.** 1 kW aik ghanta chale to 1 kWh.
MWh	Megawatt-hour	1,000 kWh. Dashboard khud switch kar leta hai jab 1,000 se zyada ho jaye.
kVA	Kilovolt-ampere	Tar aur transformer par total kitna bojh hai. Transformer ki rating isi me hoti hai.
kVAr / MVAR	Reactive power	Wo bijli jo kaam to nahi karti lekin motor aur transformer ko chahiye hoti hai.
V / kV	Volt / Kilovolt	Bijli ka "pressure". Hamare site par 11 kV (11,000 volt) aati hai.
A	Ampere	Kitni bijli beh rahi hai. Tar ki motai isi hisaab se hoti hai.
Hz	Hertz	Bijli aik second me kitni baar rukh badalti hai. 50 ke qareeb rehna zaroori hai.
%	Percent	Loading, efficiency, limit ka kitna hissa — sab me.
t	Ton	CO_2 (carbon) jo solar ki wajah se bacha.

2.3 Bijli ke system ke hissay

LAFZ	MATLAB
Substation	Wo jagah jahan high voltage aati hai aur usay kam kar ke aage bheja jata hai. Hamare paas 3 hain: PNS Hafeez, ARC, NHQ.
Transformer	Wo machine jo voltage ko high se low karti hai. Register me 109 hain, total 18,530 kVA.
Feeder	Wo tar/circuit jo substation se bijli aage buildings tak le kar jata hai. Register me 3 hain.
Busbar (Bus)	Substation ke andar mota tanba/patti jahan sab connections milte hain.

LAFZ	MATLAB
VCB (Vacuum Circuit Breaker)	Automatic switch jo kharabi hone par circuit kaat deta hai. 11 kV par 8 lage hain.
DB (Distribution Board)	Building ke andar wala board jis me fuse/breaker hote hain. Register me 475 hain.
SLD (Single Line Diagram)	Bijli ke poore system ka standard naqsha. Har electrical engineer isay parh leta hai.
HT / LT	High Tension (11 kV, mota system) aur Low Tension (400 V, building tak).
Solar PV / kWp	Solar panel jo suraj se bijli banate hain. Register me 13 system hain.

2.4 Quality ke alfaz

LAFZ	MATLAB AUR KYUN AHEM HAI
Power Factor (PF)	0 se 1 tak ka score. Batata hai ke jitni bijli aayi us ka kitna hissa asal kaam me laga. **0.95 se upar acha, 0.90 se neeche jurmana lagta hai.** Motor zyada chalein to ye neeche aata hai.
Frequency	50 Hz ke qareeb rehna chahiye (49.5 se 50.5 tak). Ye poore mulk ke grid ki halat batata hai — hum apne site se isay control nahi kar sakte.
Voltage	11 kV nominal, 10.5 se 11.5 kV tak theek hai. Kam ho to motor garam hote hain, zyada ho to insulation kharab hoti hai.
Phase Imbalance	Teen phase (R, Y, B) par bojh barabar na ho. **2% se kam rehna chahiye.** Zyada ho to motor jal jate hain — ye sabse mehnga nuqsan hai.
Losses	Wo bijli jo tar aur transformer me garmi ban kar zaya ho jati hai. Aam tor par 2–3%. Paise us ke bhi dene parte hain.
Load Factor	Average bijli ÷ sabse zyada (peak) bijli. Zyada ho to istemal barabar hai; kam ho to sirf thori der ke peak ke liye zyada paise de rahe hain.
Peak Demand	Poore arse me sabse zyada bijli jo aik waqt me li gayi. Bijli wale is par alag charge lagate hain.
Capacity	System kitna bojh utha sakta hai. Hamari total 16.7 MW hai, jabke aam istemal sirf 4.5 MW.

BAB 3

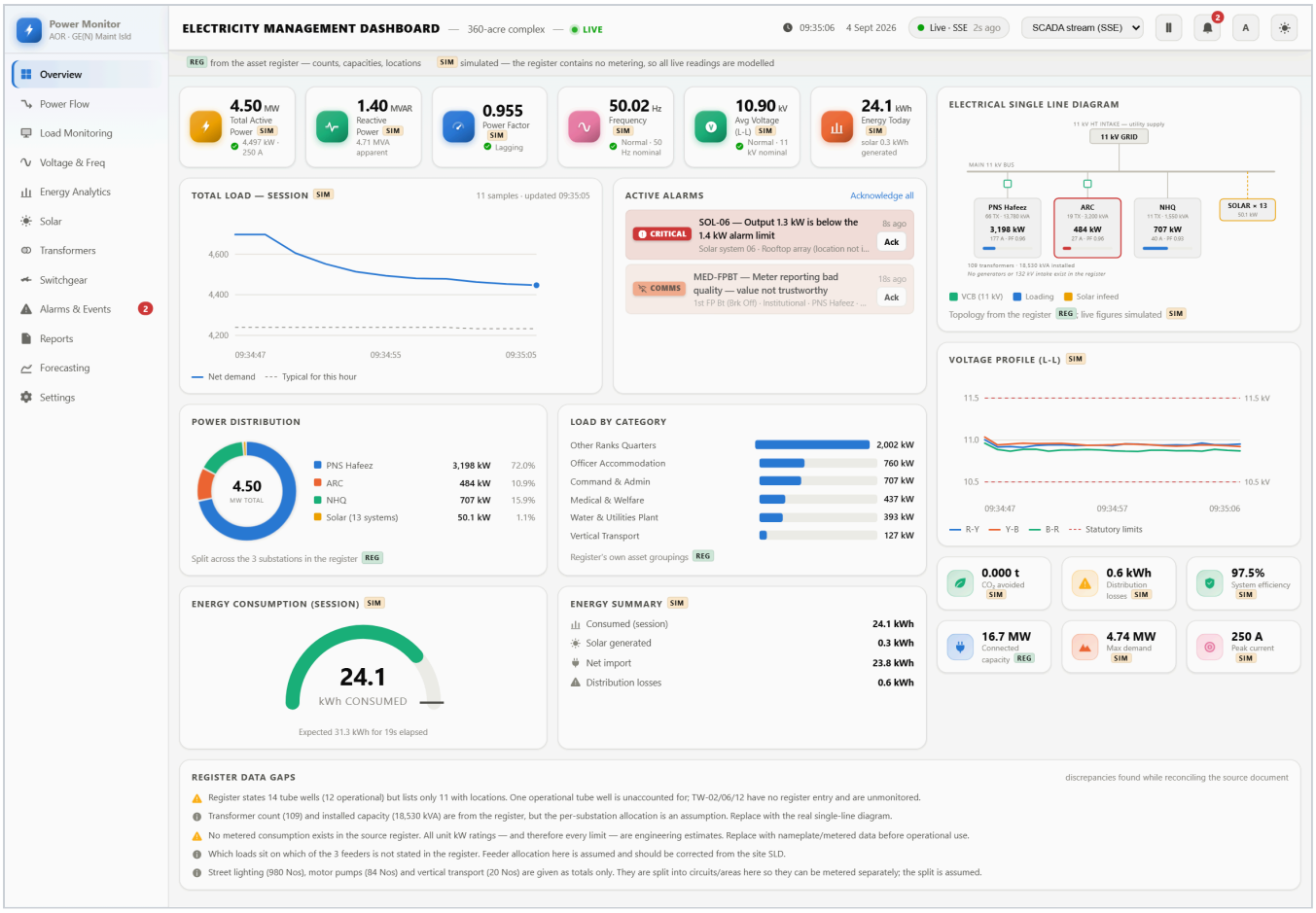
Har Tab Kya Dikhata Hai

Baayen taraf 12 tab hain. Har aik ka maqsad alag hai.

TAB	KIS SAWAL KA JAWAB DETA HAI
1. Overview	Abhi is waqt koi kharabi to nahi?
2. Power Flow	Bijli kahan se kahan ja rahi hai?
3. Load Monitoring	Har alag jagah kitni bijli le rahi hai?
4. Voltage & Freq	Jo bijli aa rahi hai wo theek hai ya kharab?
5. Energy Analytics	Ab tak kitni bijli kharch hui aur kahan?
6. Solar	Solar panel theek kaam kar rahe hain?
7. Transformers	Hamare paas kitni transformer capacity hai?
8. Switchgear	Breaker aur boards ki kya halat hai?
9. Alarms & Events	Kya kharabi hui aur kya usay handle kiya gaya?
10. Reports	Total figures kya hain, Excel me export ho sakta hai?
11. Forecasting	Agle 24 ghante me bijli ka load kitna hoga?
12. Settings	Har number kahan se aaya?

Table 3.1 — 12 tabs aik nazar me.

3.1 Overview — sabse ahem screen



Tasveer 1 — Overview screen. Ye wo screen hai jo deewar par lagi TV/monitor par hamesha chalti rehni chahiye.

Is par ye cheezen hain:

HISSA	KYA DIKHATA HAI
6 bare numbers (KPI)	Total Power (MW), Reactive Power (MVAR), Power Factor, Frequency (Hz), Voltage (kV), Energy (kWh). Har aik ke neeche hara tick ya peela warning hota hai.
Total Load chart	Waqt ke saath bijli ka graph. Neeche dotted line "typical" hai — yaani is waqt aam tor par kitni honi chahiye.
Active Alarms	Jo jagahen abhi limit se upar hain. Har aik ke saath "Ack" ka button.
Power Distribution	Gol chart — 3 substation aur solar me se kaun kitna de raha hai.
Load by Category	Kaun si qism ke assets sabse zyada bijli le rahe hain.
Energy gauge	Ab tak kitni energy kharch hui, aur kitni honi chahiye thi.
Energy Summary	Kharch, solar se bani, grid se li, aur losses.
Single Line Diagram	Poore system ka naqsha — grid, bus, 3 substation, solar.
Voltage Profile	Teenon phase ka voltage graph, upar neeche laal limit lines ke saath.
6 chhote tiles	CO_2, losses, efficiency, capacity, max demand, current.
Register Data Gaps	Document me jo kami/ghalti mili, wo saaf saaf likhi hui.

3.2 Baqi tabs — mukhtasar

2. Power Flow

Aik naqsha jis me moti lakeeren dikhati hain ke bijli kis raste kitni ja rahi hai. Jitni moti lakeer, utni zyada bijli. Char column: source → bus → substation → group. Neeche har feeder kitna load utha raha hai (percent me).

3. Load Monitoring

Live consumption for every monitored location, with its warning and alarm limits.

Tasveer 2 — Points table. Har jagah ki live reading, typical, limit aur status.

Sari 110 jagahon ki list. Columns ka matlab:

COLUMN	MATLAB
Tag	Us jagah ka code (jaise TW-01).
Location	Naam aur jagah.
Now	Abhi kitni bijli le raha hai (kW).
Typical	Is waqt aam tor par kitni leni chahiye.
vs typical	Normal se kitna kam/zyada (percent).
Limit	Alarm ki hadd.
% of limit	Limit ka kitna hissa istemal ho chuka. **Ye sabse kaam ka column hai.**
Trend	Chhota sa graph — pichle kuch minute ka.
Status	Normal / Warning / Over limit / Offline.
Updated	Aakhri reading kitni der pehle aayi.

Upar search box hai, group aur substation ka filter hai, aur aik button "Only points needing attention" — is se sirf wo jagahen dikhti hain jinme masla hai. Kisi bhi row par click karein to us jagah ki poori tafseel khul jati hai.

4. Voltage & Frequency

Frequency ka graph (49.5–50.5 Hz ki limit ke saath), teen substation ke alag alag voltage graph, aur aik table jis me har substation ka voltage, imbalance, current aur power factor hai. **Sabse zaroori number: Phase Imbalance** — 2% se kam rehna chahiye.

5. Energy Analytics

Kitni energy kharch hui, average kitna load raha, peak kab tha, load factor kitna hai. Aik gol chart me: grid se li gayi, solar se bani, aur losses. Aik khaas graph **"Load Duration Curve"** — ye batata hai ke kitne waqt tak peak ke qareeb rahe. Neeche top 10 sabse zyada kharch karne wali jagahen.

6. Solar

13 solar system ki performance. Sabse ahem number **"Performance Ratio"** hai — yaani jitni bijli abhi ke suraj me banni chahiye thi, us ka kitna percent asal me bani. Subah kam output normal hai; dopahar me kam output kharabi hai. Sirf ratio se ye farq pata chalta hai.

7. Transformers

Ye akela tab hai jis par kuch bhi banaya hua nahi hai — sab kuch register se hai. 109 transformer, 18,530 kVA, aur har substation par kitne lage hain.

8. Switchgear

8 breakers (VCB) ki halat: Closed (chal raha hai), Standby (spare), Open (band kiya gaya), Tripped (kharabi par khud band ho gaya). Neeche 475 distribution boards ka summary.

9. Alarms & Events

Sare active alarm, aur poori history. Har alarm ke teen waqt likhe hote hain: **kab hua, kab dekha gaya, kab theek hua**. Ye teen waqt kisi bhi incident ke baad sabse zaroori sawal ka jawab dete hain.

10. Reports

Session ka summary aur do CSV export buttons (Excel me khul jate hain): aik sari jagahon ka, aik alarm history ka.

11. Forecasting

Agle 24 ghante ka andaza. Solid line = jo ho chuka, dotted line = jo hone ka andaza hai. Shaded (halka) hissa confidence range hai — aage jaa kar chaura hota jata hai kyunki door ka andaza kam pakka hota hai.

> **FORECAST KE BAARE ME SACH**
>
> Ye asli "trained forecast" nahi hai. Ye sirf har group ke rozana ke pattern ko jorr kar banaya gaya andaza hai, kyunki purana record maujood hi nahi hai jis se model seekhta. Jab kuch hafton ka asli data jama ho jaye ga, tab is ko proper model se badla ja sakta hai. Ye baat dashboard par bhi likhi hui hai.

12. Settings

Poori tafseel ke kaun sa number kahan se aaya. Agar koi banda kisi figure par sawal uthaye to ye tab kholein.

BAB 4

Alarm System Kaise Kaam Karta Hai

Ye dashboard ka sabse ahem hissa hai, aur is me kaafi soch bichar ki gayi hai.

4.1 Masla kya hota hai

Agar alarm ka system saada bana diya jaye — yaani "jab number limit se upar jaye to alarm baja do" — to aik hi jagah se sainkron alarm aate hain aur operator alarm dekhna hi chhor deta hai. Isay **alarm flood** kehte hain, aur ye control room ka sabse aam masla hai.

4.2 Is system me kya kya bandobast hai

CHEEZ	VALUE	KYA KARTA HAI
On-delay	6 second	Limit 6 second tak lagatar cross ho, tabhi alarm bajta hai. Aik jhatka alarm nahi de sakta.
Hysteresis	4%	Alarm khatam karne ke liye value ko limit se *4% neeche* aana parta hai. Warna value limit par baithi rahe to alarm baar baar on-off hota rahega.
Off-delay	12 second	12 second tak theek rahe, tabhi alarm khatam hota hai.
Latching	—	Alarm khud se gayab nahi hota. Jab tak koi banda "Ack" na kare, screen par rehta hai.
Ringback	—	Agar kharabi khud theek ho gayi lekin kisi ne dekha nahi, tab bhi alarm list me rehta hai. Warna aane jane wali kharabi kabhi pakri hi na jaye.
Escalation	—	Agar wahi masla aur bigar jaye to naya alarm nahi banta — wahi purana alarm update ho jata hai.
Re-arm	90 second	Aik alarm khatam hone ke baad wahi jagah 90 second tak dobara wahi alarm nahi de sakti. Aik kharab sensor se darjnon entries nahi banti.

4.3 Teen level ke alarm

STATUS	MATLAB
Normal	Sab theek hai, kuch karne ki zaroorat nahi.
Warning	Warning limit se upar chala gaya. Dekh lena chahiye, jaldi nahi.
Over limit	Asli limit cross ho gayi. Abhi tawajjo chahiye.
Offline	Meter se koi reading nahi aa rahi. **Ye bijli ka masla nahi, connection ka masla hai.**
Not monitored	Register me hai lekin meter laga hi nahi (jaise TW-02, TW-06, TW-12).

> **AIK AHEM BAAT — "OFFLINE" AUR "0 KW" ALAG CHEEZEN HAIN**
>
> Agar meter se rabta toot jaye to dashboard usay **"Offline"** dikhata hai, **"0 kW" nahi**. Kyunki inka matlab bilkul alag hai: aik ka matlab hai connection kharab hai, doosre ka matlab hai bijli waqai band hai. Bohot se systems is me ghalti karte hain.

4.4 Har status me rang ke saath icon aur lafz bhi hota hai

Sirf rang par bharosa nahi kiya gaya, kyunki (a) kuch logon ko rang ki pehchan me mushkil hoti hai, (b) print black-and-white me nikle to rang khatam ho jata hai, aur (c) jaldi me lafz parhna rang se zyada tez hota hai.

BAB 5

Data Kahan Se Aata Hai

Kaun sa number asli hai, kaun sa andaza, aur kaun sa banaya hua.

CHEEZ	HALAT	TAFSEEL
Assets ki tadaad	**Asli**	2,157 buildings, 980 lights, 84 pumps, 14 tube wells, 475 DB, 13 solar — sab register se.
Transformers	**Asli**	109 unit, 18,530 kVA. Register ke total se bilkul match karta hai.
Jagahon ke naam	**Asli**	Tube wells, filter plants, water tanks — naam aur jagah register se.
System ka structure	**Asli**	3 substation, 3 feeder, 8 VCB, 38 km HT, 108 km LT tar.
Har unit ki kW rating	**Andaza**	Register me nahi hai. Engineering andaza lagaya gaya. **Sari limits isi par bani hain.**
Kaun sa load kis feeder par	**Farz kiya**	Register me nahi likha. Site ke asli SLD se theek karna hoga.
Sari live readings	**Banai hui**	Computer program se generate hoti hain.
Voltage, current, PF, Hz	**Banai hui**	Power se hisaab laga kar nikali gayi.
Breaker ki halat	**Banai hui**	Breaker ki tadaad aur jagah asli hai, halat nahi.
Purana record (history)	**Kuch nahi**	Page refresh karne par sab khatam. Koi database nahi hai.

> SAAF BAAT
>
> Is waqt dashboard par **koi bhi live number asli meter se nahi aa raha**. Ye aik mukammal, sahi kaam karne wala monitoring system hai jo abhi banaye gaye data par chal raha hai. Software taiyar hai — sirf asli meter lagne baqi hain.

5.1 Asli SCADA se kaise joreinge

```
Asli meters (Modbus)
      ↓
  Gateway / bridge service      ← ye banana parega
      ↓  (JSON data)
  /api/stream
      ↓
  Dashboard  — koi tabdeeli nahi
```

Poore dashboard me sirf **aik file** badalni parti hai (`src/source.js`) jis me batana hota hai ke data kahan se lena hai. Baqi sab screens khud ba khud asli data dikhane lag jayengi.

> DO BAATEIN SCADA JORTE WAQT ZAROOR YAAD RAKHEIN
>
> **1.** Agar kisi meter se rabta na ho to us ka data *bhejein hi nahi* — us ki jagah **0 mat bhejein**. Warna "connection kharab" aur "bijli band" ka farq khatam ho jayega.
>
> **2.** Waqt (timestamp) wo bhejein jo *meter par* reading li gayi thi, na ke jab data pohancha. Warna atka hua meter bhi theek lagta rahega.

BAB 6

Dashboard Kaise Parhein

Rozana ka tareeqa, aur akhir me zaroori baatein.

6.1 Rozana ka 4-step check (10 second)

1. **Overview ke 6 bare numbers dekhein.** Agar Power Factor, Frequency aur Voltage teenon par hara tick hai, to bijli ki quality theek hai.
2. **Active Alarms dekhein.** Khali hai to koi jagah limit se upar nahi.
3. **Load ka graph dekhein.** Solid line dotted line ke qareeb chal rahi hai?
4. **Upar connection pill dekhein.** "Live" likha ho. Agar "Paused" ya "Feed lost" hai to upar wale teen check bekaar hain — data purana hai.

Ye chaar theek hain to aur kuch dekhne ki zaroorat nahi.

6.2 Agar kuch masla ho

STEP	KAHAN JAYEIN	KYUN
1	Load Monitoring	"Only points needing attention" button dabayein — sirf masle wali jagahen dikhengi.
2	Us row par click	Us jagah ka graph limits ke saath, aur uske purane alarm.
3	Voltage & Freq	Ye dekhne ke liye ke masla load ka hai ya aane wali bijli ka.
4	Power Flow / Switchgear	Masla kis feeder ya breaker par hai.
5	Reports	Record ke liye CSV nikal lein (Excel me khulta hai).

6.3 Ahem baatein jo zehen me rakhni chahiye

BAAT	TAFSEEL
Sari limits andazay par hain	Kisi unit ki asli kW rating register me nahi thi. Jab tak asli nameplate ya meter data na aaye, limits sirf "provisional" hain. **Ye sabse zaroori baat hai.**
Record save nahi hota	Page refresh karne par sara data aur alarm history khatam. Zaroori ho to pehle CSV export kar lein.
Login nahi hai	Jo bhi link khole, poora access mil jata hai, alarm Ack karne ka bhi. Alarm log me sirf "operator" likha jata hai, naam nahi.
Register me aik ghalti mili	Document kehta hai **14 tube wells, 12 chalu**, lekin naam sirf **11** ke likhe hain. Aik chalu tube well ka hisaab nahi milta. Ye facilities team se poochna chahiye.
Generator nahi hain	Register me kisi generator ka zikr nahi, isliye diagram me bhi nahi dikhaya gaya. Sirf 11 kV grid aur solar.

6.4 Aage kya karna hai

Is dashboard ko asli system banane ke liye teen cheezen chahiye:

1. **Asli meters** — shuru me sirf 3 substation par lagayein. Isi se poora Overview, Voltage aur Power Flow asli ho jayega.
2. **Asli kW ratings** — taake limits qabil-e-bharosa hon.
3. **Database (historian)** — taake record mehfooz rahe aur proper forecasting ho sake.

AAKHRI BAAT

Software ke lehaz se ye system mukammal aur professional hai. Kami sirf data ki hai — aur wo software ka masla nahi, meter lagane ka kaam hai. Achi baat ye hai ke dashboard khud saaf saaf batata hai ke kaun sa number asli hai aur kaun sa nahi.